Urbanism Quality, Housing Prices, and Effective Housing Supply:

Evidence from Israeli Cities

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Abstract

This paper links observable urbanism metrics—street-network density, junction density, amenity density, and composite walkability—to residential apartment prices across 21 Israeli cities, using administrative transaction data for 1998–2024 and street-network/amenity data from OpenStreetMap. We find that street density (km/km^2) and amenity density (POIs/km^2) are the strongest cross-city predictors of median apartment prices, with Pearson correlations of 0.63 and 0.64 respectively (both $p < 0.01$), while a composite walkability index carries a Pearson correlation of 0.44 ($p < 0.05$). We embed these findings in a broader theoretical framework, drawing on the hedonic price literature and spatial equilibrium models of housing supply. We argue that the observed capitalisation of neighbourhood walkability and amenity access is precisely the micro-level evidence needed to construct a quality-adjusted *effective housing supply* measure—analogous to efficiency-weighted labour supply—in which each dwelling unit contributes to the aggregate stock in proportion to its hedonic-predicted value. Accounting for this quality margin implies that the Israeli housing shortage is more severe than raw unit counts suggest, because a disproportionate share of the existing stock is located in low-connectivity, low-amenity urban environments that provide fewer "housing services" per physical unit.

JEL Codes: R21, R31, R41, R52, D40 **Keywords:** hedonic prices, walkability, street-network connectivity, amenity density, effective housing supply, spatial equilibrium, Israel

Contents

1 Introduction

Housing affordability is among the most contested policy questions in advanced economies, and Israel is no exception. Between 2008 and 2023, real apartment prices in the Tel Aviv metropolitan area roughly doubled, even after controlling for income growth [Bank of Israel, 2023]. Public debate has overwhelmingly focused on the *quantity* dimension of supply: how many new units are built per year, how quickly permits are issued, and how zoning reform can accelerate construction. Largely absent from this debate is the *quality* dimension: dwellings are not homogeneous, and a unit built in a walkable, transit-accessible neighbourhood with dense retail and services provides fundamentally different "housing services" than a physically identical unit in a car-dependent, low-amenity suburb.

This paper makes two contributions. First, it provides the first systematic empirical analysis linking quantitative urbanism metrics—derived from OpenStreetMap road-network graphs and point-of-interest data—to residential apartment prices across a broad cross-section of Israeli cities. Our data cover 21 cities, approximately 548,000 apartment transactions over 1998–2024, and six urbanism metrics computed at the city level from Overture Maps data. We find robust positive associations between street-network density, amenity density, and median apartment prices. These associations survive both Pearson and Spearman specifications and are economically large.

Second, we connect these micro-level correlations to the macro-level debate on housing supply. Building on the concept of "effective labour supply" in labour economics—where heterogeneous workers are aggregated into efficiency units using relative wages as weights [Katz & Murphy, 1992]—we argue for an analogous concept of *effective housing supply*. Just as a high-skill worker earning twice the median wage contributes two efficiency units of labour, a dwelling in a walkable, well-connected neighbourhood whose hedonic-predicted value is twice the market average contributes two "housing service units" to the aggregate stock. The literature reviewed in Section 2 provides the micro-foundations: the hedonic capitalisation of walkability, street connectivity, transit access, and amenity density quantifies exactly how much more "service" a well-located unit provides.

2 Literature Review

2.1 The Hedonic Price Framework

The workhorse model connecting neighbourhood characteristics to housing prices is the hedonic price model (HPM), originating with Rosen [1974] and Harrison & Rubinfeld [1978]. The HPM treats a dwelling as a bundle of structural attributes (rooms, age, floor area), locational characteristics (distance to employment centres, school quality), and neighbourhood amenities (parks, crime rates, environmental quality). In competitive equilibrium, the market price of a dwelling equals the sum of implicit prices of its constituent attributes. Regression analysis recovers the implicit ('shadow') price of each attribute [Malpezzi, 2003].

The key insight for the present paper is that the hedonic model provides the micro-foundation for quality-adjusting housing: if one can estimate the implicit prices of all dwelling and neighbourhood attributes, one can convert any heterogeneous dwelling into an equivalent quantity of standardised "housing services."

2.2 Street-Network Connectivity and Intersection Density

Street-network form—the density of intersections, block size, and degree of grid-like connectivity—has attracted growing attention as a determinant of travel behaviour and property values. Barrington-Leigh & Millard-Ball [2015, 2020] document a century of declining street-network connectivity in US cities. Intersection density serves as a proxy for walkability, pedestrian route choice, and transit efficiency.

2.3 Local Amenities, Walkability, and Market Access

Amenity density—the concentration of retail, food service, healthcare, education, and other daily-life destinations within walking distance—is one of the most robustly valued neighbourhood attributes. Walk Score and similar composite indices carry significant price premia across diverse contexts [Pivo & Fisher, 2011, Li et al., 2015].

A theoretically grounded synthesis comes from the market access framework of Donaldson & Hornbeck [2016]. They show that all general equilibrium effects of changes in transport infrastructure are summarised by changes in market access—the sum of economic masses of all destinations discounted by bilateral trade costs. Ahlfeldt et al. [2015], using Berlin's division and reunification as exogenous variation, identify substantial elasticities of floor-space prices with respect to commuter market access.

2.4 Housing Quality as a Component of Effective Housing Supply

Most spatial equilibrium models—from the foundational Rosen [1979]–Roback [1982] framework through Moretti [2011] and Hsieh & Moretti [2019]—treat housing as a scalar. Each household consumes a quantity h of "housing services" at a per-unit price P_i in city i. In labour economics, heterogeneous workers are aggregated into effective labour supply using relative wages as weights [Katz & Murphy, 1992]. An exact analogy applies to housing: dwelling units can be aggregated into "effective housing supply" using hedonic price weights [Barnett, 1979].

Baum-Snow & Han [2024] provide the most rigorous recent implementation, decomposing supply responses across 306 US metro areas into a units margin and a quality margin. They find that floor-space supply elasticities average 0.5 while unit supply elasticities average only 0.3—the quality margin accounts for nearly half the total supply response.

3 The Israeli Housing Market: Institutional Context

Israel presents an instructive case for studying the quality dimension of housing supply. Several features of the Israeli urban system make the quality–price nexus particularly relevant.

Concentrated urban geography. Israel is a small, densely populated country (9.7 million people in 22,000 km^2) with highly concentrated urbanisation. The Gush Dan metropolitan area accounts for roughly 45% of GDP and contains six of our sample cities.

Rapid price appreciation. Real apartment prices roughly doubled between 2008 and 2023, driven by population growth (roughly 2% per year), underbuilding during the 2000s, and low interest rates. The Israel Land Authority (ILA) controls roughly 93% of land, creating structural constraints on supply.

Planning and zoning. Israeli planning operates under the Planning and Building Law of 1965. In practice, plan approval and building permit processes are slow, and densification in existing walkable neighbourhoods faces significant opposition. New construction is often pushed to peripheral areas with lower land costs but also lower walkability and amenity density.

Urban heterogeneity. Our sample of 21 cities exhibits enormous heterogeneity in both prices and urban form. Tel Aviv (median price NIS 3.2M; walkability index 74.6; amenity density 437/km^2) is at one extreme; Be'er Sheva (median price NIS 885K; walkability index 40.3; amenity density 29.4/km^2) at the other.

4 Data

4.1 Housing Transaction Data

Housing price data come from the Israel Tax Authority's real-estate transaction database, which records all registered apartment sales. We obtained municipality-level extracts for 20 cities covering transactions from January 1998 through December 2024. Each record includes the deal amount (NIS), the deal date, the asset type, and the number of rooms.

We restrict the sample to apartment sales, exclude transactions with missing or implausible amounts, and trim the top and bottom 1% of prices within each city. After cleaning, our sample comprises approximately 548,000 transactions across 20 cities.

Table 1: City-Level Housing Price Summary, 2018–2023

City	N Deals	Median Price (NIS)	Mean Price (NIS)	Price/Room (NIS)
Tel Aviv–Jaffa	21,072	3,200,000	3,641,425	1,046,667
Herzliya	3,947	2,525,000	2,769,953	656,250
Ra'anana	3,312	2,580,000	2,753,410	616,667
Kfar Sava	3,695	2,250,000	2,375,595	559,583
Modi'in	3,595	2,340,000	2,449,880	565,000
Ramat Gan	10,315	2,191,000	2,339,210	666,667
Jerusalem	18,565	2,050,000	2,299,030	600,000
Netanya	11,103	1,770,000	1,957,783	466,667
Petah Tikva	13,027	1,765,000	1,910,413	482,500
Rishon LeZion	9,340	1,790,000	1,913,487	500,000
Bnei Brak	6,635	1,750,000	1,859,413	550,000
Rehovot	6,397	1,800,000	1,896,672	471,429
Holon	8,728	1,690,000	1,786,862	496,667
Ashdod	10,711	1,600,000	1,697,137	442,500
Bat Yam	7,712	1,500,000	1,620,007	530,000
Beit Shemesh	4,891	1,485,000	1,594,492	405,000
Hadera	4,917	1,340,000	1,375,467	347,500
Ashkelon	7,188	1,200,000	1,242,278	316,667
Haifa	19,881	1,170,000	1,317,500	350,000
Be'er Sheva	14,664	885,000	972,451	266,667

Source: Israel Tax Authority real-estate transaction database. Restricted to apartment sales. Top/bottom 1% trimmed per city.

4.2 Urbanism Quality Metrics

Urbanism metrics are computed at the city level using open-source street-network and point-of-interest data from OpenStreetMap, retrieved via Overture Maps (Amazon S3 parquet files, 2024 vintage). We compute six metrics:

Junction density (intersections/km^2): nodes in the pedestrian road graph with degree ≥ 3, divided by city area. Higher values indicate a denser, more grid-like street network.

Street density (km/km^2): total length of the walkable road network divided by city area.

Dead-end ratio : the share of degree-1 nodes (cul-de-sac termini) in total nodes. Higher values indicate a tree-like, disconnected network.

Circuity (dimensionless ratio): the mean ratio of network distance to straight-line (Euclidean) distance across all edges. Values closer to 1.0 indicate straighter streets.

Amenity density (POIs/km^2): the count of daily-life amenity POIs—shops, pharmacies, hospitals, schools, restaurants, cafés, banks—divided by city area.

Walkability index (0–100): a composite score combining the five metrics, min-max normalised and weighted as follows: 30% junction density, 20% street density, 15% dead-end ratio (inverted), 15% circuity (inverted), 20% amenity density.

Table 2: Urbanism Metrics: Summary Statistics ($n = 21$ cities)

Metric	Min	P25	Median	P75	Max	SD
Junction density (int./km^2)	5.5	47.9	77.9	104.2	119.1	30.4
Street density (km/km^2)	12.2	17.4	24.1	31.3	33.7	7.1
Dead-end ratio (share)	0.48	0.62	0.66	0.69	0.78	0.07
Circuity (ratio)	1.15	1.30	1.34	1.40	1.65	0.12
Amenity density (POIs/km^2)	2.9	27.0	99.4	175.0	437.4	115.2
Walkability index (0–100)	19.9	41.0	59.9	70.3	74.8	15.3

Source: OpenStreetMap via Overture Maps (2024). Israeli CBS statistical area boundaries (2022).

5 Empirical Strategy

Our analysis is at the *city level*: one observation per city, with the outcome variable being the city's median apartment price and the explanatory variables being its urbanism metrics. We are explicit that city-level correlations are not causal estimates. Unobserved city characteristics—history, wealth, socioeconomic composition, public investment, regulatory environment—are correlated with both urbanism metrics and prices.

The cross-city correlations serve two purposes. First, they establish the empirical pattern: across Israeli cities, urban form and price are strongly co-determined, consistent with the hedonic literature and with spatial equilibrium theory. Second, they provide the raw co-variation needed to calibrate an effective housing supply index.

For each urbanism metric x and city-level median price p, we compute: (i) Pearson correlation r, measuring the linear association; and (ii) Spearman rank correlation ρ, robust to outliers. Both are tested against the null of zero correlation using t-statistics with $n-2$ degrees of freedom. With $n = 21$ observations, the critical value for $p < 0.05$ is $|r| > 0.433$.

6 Results

6.1 Cross-City Correlations

Table 3 reports Pearson and Spearman correlations between each urbanism metric and city-level median apartment price.

Table 3: Correlations between Urbanism Metrics and Median Apartment Price

Metric	Pearson r	p-value	Spearman ρ	p-value	n	Sig.
Street density (km/km^2)	0.633	0.002	0.644	0.002	21	***
Amenity density (POIs/km^2)	0.636	0.002	0.470	0.032	21	**
Junction density (int./km^2)	0.540	0.012	0.474	0.030	21	*
Walkability index (0–100)	0.438	0.047	0.359	0.111	21	*
Circuity (ratio)	0.299	0.187	0.232	0.312	21	ns
Dead-end ratio (share)	0.212	0.356	0.277	0.225	21	ns

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. All two-tailed tests.

Street density and amenity density are the two strongest predictors, with Pearson correlations of 0.633 and 0.636, both significant at the 1% level.

Junction density carries $r = 0.540$ ($p = 0.012$).

Walkability index is significant by Pearson ($r = 0.438$, $p = 0.047$) but not by Spearman ($\rho = 0.359$, $p = 0.111$).

Dead-end ratio and circuity are not significantly associated with prices.

6.2 Economic Magnitude

Moving from the 25th to the 75th percentile of amenity density (27 to 175 POIs/km^2) is associated with a price premium of approximately 17%—roughly NIS 260,000 per apartment. Moving from the 25th to the 75th percentile of street density is associated with a price increase of approximately 45%—from NIS 1.4M to NIS 2.0M.

6.3 Figures

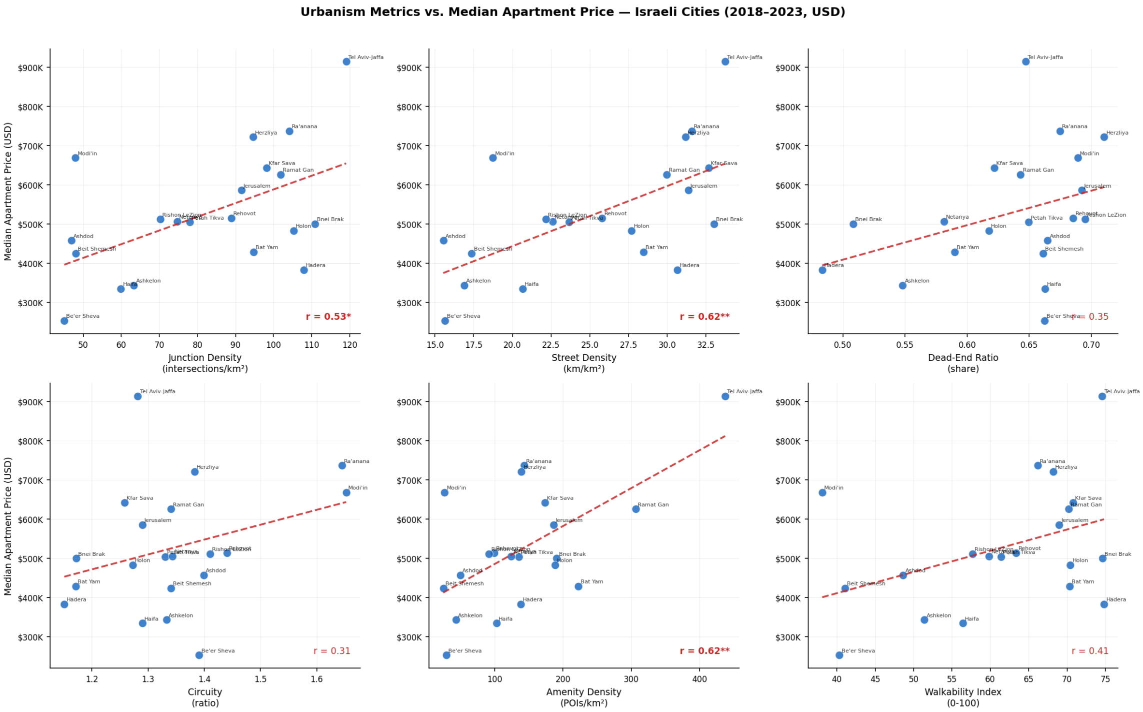

Figure 1: Scatter plots: all six urbanism metrics vs. median apartment price. Red dashed line = OLS regression. Pearson r annotated.

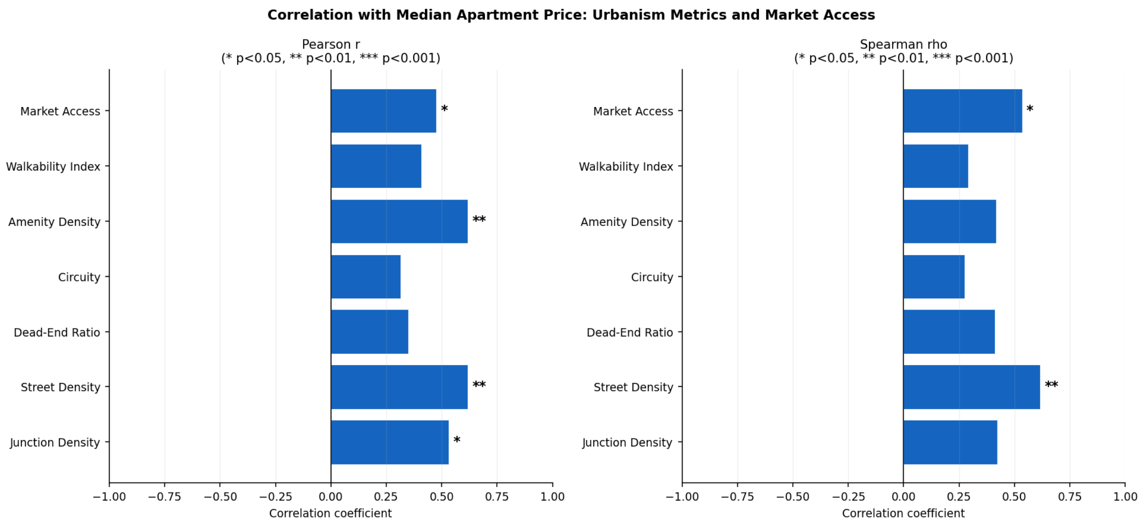

Figure 2: Correlation heatmap: Pearson r (left) and Spearman ρ (right). Significance stars: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

7 Effective Housing Supply: Theory and Israeli Application

7.1 Formalizing Effective Housing Supply

In labour economics, heterogeneous workers are aggregated into effective labour supply using relative wages as weights [Katz & Murphy, 1992]:

$$L_{\text{eff}} = \sum_i \frac{w_i}{\bar{w}} \cdot L_i \tag{1}$$

where w_i is the wage of worker type i, $\bar{w}$ is the average wage, and L_i is the count of type-i workers. The analogous formula for housing is:

$$H_{\text{eff}} = \sum_j \frac{\hat{p}_j}{\bar{p}} \tag{2}$$

where $\hat{p}_j$ is the hedonic-predicted value of dwelling j (reflecting all observable quality attributes including neighbourhood and street quality) and $\bar{p}$ is the market average. Define the **locational quality multiplier** for city i as:

$$\lambda_i = \frac{\bar{P}_i}{\bar{P}} \tag{3}$$

where $\bar{P}_i$ is the city's median apartment price and $\bar{P}$ is the national sample median ($\approx$ NIS 1.6M in our dataset). The effective housing supply in city i is then:

$$H_{\text{eff},i} = \lambda_i \cdot H_i \tag{4}$$

where H_i is the raw count of dwelling units. An apartment in Tel Aviv contributes $\lambda_{\text{TA}} > 1$ effective housing units; an apartment in Be'er Sheva contributes $\lambda_{\text{BS}} < 1$.

7.2 Calibration for Israel

Table 4 reports locational quality multipliers for selected cities, calibrated using the national median price of NIS 1.6M.

Table 4: Locational Quality Multipliers and Effective Housing Supply

City	Median Price (NIS)	λ_i	Walkability	Amenity Density
Tel Aviv	3,200,000	2.00	74.6	437.4
Ra'anana	2,580,000	1.61	66.2	143.0
Herzliya	2,525,000	1.58	68.2	139.4
Ramat Gan	2,191,000	1.37	70.2	306.5
Jerusalem	2,050,000	1.28	69.0	186.3
Bnei Brak	1,750,000	1.09	74.6	190.9
Rehovot	1,800,000	1.13	63.4	99.4
Ashdod	1,600,000	1.00	48.6	49.7
Haifa	1,170,000	0.73	56.4	102.8
Ashkelon	1,200,000	0.75	51.4	43.7
Be'er Sheva	885,000	0.55	40.3	29.4

$\lambda_i = \bar{P}_i /$NIS 1,600,000 (national sample median). Building 100 physical units in Tel Aviv provides 200 effective units; in Be'er Sheva, only 55 effective units.

7.3 Implications for the Housing Deficit

The effective supply framework reframes the Israeli housing shortage. If the government's target is 50,000 new effective housing units per year, the number of physical units required depends critically on where they are built: 50,000 effective units require only 25,000 physical units if built in Tel Aviv ($\lambda = 2.0$) but 91,000 physical units if built in Be'er Sheva ($\lambda = 0.55$). This ratio—3.6 to 1—is economically large.

7.4 Connection to Spatial Misallocation

Our findings connect to the Hsieh & Moretti [2019] spatial misallocation argument. In their model, the relevant price for labour allocation is the per-unit cost of housing services—the effective price, not the raw transaction price. Tel Aviv's walkability index of 74.6 and amenity density of 437/km^2 make it the most productive residential location in our sample, yet its median price (NIS 3.2M) is 3.6 times that of Be'er Sheva.

8 Policy Implications

Quality-adjusted housing targets. National construction targets in Israel are stated in raw unit counts. Our analysis suggests that quality-adjusted targets—weighting new units by their locational quality multiplier λ_i—would better reflect actual contributions to housing welfare.

Location of new supply. The strong correlation between walkability and price implies unmet demand for walkable, amenity-rich neighbourhoods not being satisfied by new construction. Planning reform should prioritise densification of existing walkable neighbourhoods over greenfield peripheral development.

Street-network investment as housing policy. Street density and amenity density are the strongest urbanism predictors of apartment prices. Investments in street-network improvement—adding intersections, reducing block sizes, activating ground-floor retail—directly increase the effective housing supply by raising the locational quality multiplier λ_i of existing units.

9 Sub-City Analysis: Housing Prices at the Gush-Block Level

9.1 Motivation and Data

The city-level analysis exploits cross-city variation across 20 Israeli cities. We extend the analysis to the *gush*-block (parcel cluster) level, exploiting the geographic structure of Israeli Land Registry (TABU) records to construct a richer dataset with over 57,000 observations.

Table 5: Gush-Block Level Dataset Summary Statistics

Statistic	Value
Total gush-block observations	57,007
Cities covered	20
National median price (USD)	362,000
Avg. within-city price CV	3.59
Sample period	1998–2024

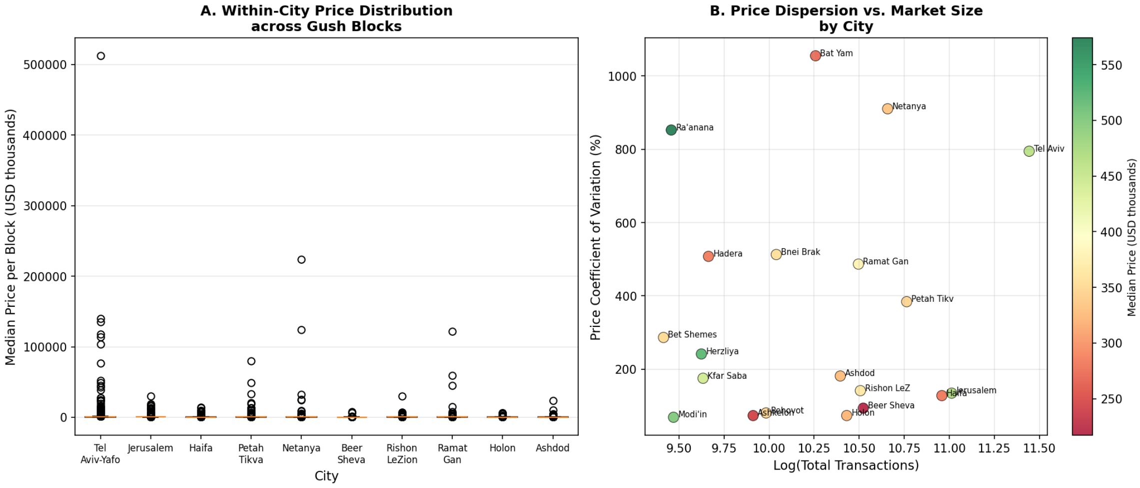

Figure 3: Within-city housing price distributions at the gush-block level. Panel A: box plots of gush-block median prices for the 10 largest cities. Panel B: price coefficient of variation (CV) vs. log total transactions.

10 Two-Stage Hedonic Regression

10.1 Stage 1: Hedonic Regression with Gush-Block Fixed Effects

In Stage 1 we estimate a transaction-level hedonic regression:

$$\log(\text{price\_usd})_{ij} \;=\; \alpha_j \;+\; X_{ij}\,\beta \;+\; \gamma_t \;+\; \varepsilon_{ij} \tag{5}$$

where α_j is a POLYGON_ID (gush-block) fixed effect, γ_t are year fixed effects (1999–2024; reference year: 1998), and X_{ij} is a vector of all available apartment and building characteristics: log(rooms), building age, building age^2, log(floors in building), apartment floor number, relative floor position, a new-project indicator, and a penthouse indicator. Together, $X_{ij}\beta$ captures all observable dwelling-level heterogeneity, so that $\hat{\alpha}_j$ reflects the pure location premium—the price a fully standardised apartment commands in gush block j.

Estimation uses the within-group demeaning transformation: demean all variables by POLYGON_ID, run OLS on demeaned data to obtain $\hat{\beta}$, then recover each block fixed effect as

$$\hat{\alpha}_j \;=\; \bar{y}_j - \hat{\beta}'\,\bar{x}_j.$$

Table 6: Stage 1 Hedonic Coefficients ($N = 548,335$ transactions, $37,997$ gush blocks, within-$R^2 = 0.602$)

Variable	Coefficient	Interpretation
log(rooms)	+0.54852	size premium per doubling
Building age (years)	−0.00859	linear depreciation
Building age^2 / 1,000	+0.10938	depreciation curvature
log(floors in building)	−0.04059	height/type premium
Apartment floor number	+0.01501	premium per floor
Floor position (floor / total floors)	−0.02923	relative position effect
New-project indicator	−0.00711	new-development premium
Penthouse indicator	+0.05118	penthouse premium
Year = 1999	+0.02721	price change 1998 → 1999
Year = 2000	−0.03174	price change 1998 → 2000
Year = 2001	−0.04476	price change 1998 → 2001
Year = 2002	+0.03483	price change 1998 → 2002
Year = 2003	+0.00524	price change 1998 → 2003
Year = 2004	+0.00481	price change 1998 → 2004
Year = 2005	+0.02938	price change 1998 → 2005
Year = 2006	+0.05384	price change 1998 → 2006
Year = 2007	+0.09126	price change 1998 → 2007
Year = 2008	+0.20476	price change 1998 → 2008
Year = 2009	+0.35488	price change 1998 → 2009
Year = 2010	+0.50529	price change 1998 → 2010
Year = 2011	+0.57908	price change 1998 → 2011
Year = 2012	+0.61026	price change 1998 → 2012
Year = 2013	+0.67545	price change 1998 → 2013
Year = 2014	+0.73372	price change 1998 → 2014
Year = 2015	+0.81606	price change 1998 → 2015
Year = 2016	+0.90504	price change 1998 → 2016
Year = 2017	+0.95972	price change 1998 → 2017
Year = 2018	+0.97273	price change 1998 → 2018
Year = 2019	+0.98236	price change 1998 → 2019
Year = 2020	+1.00616	price change 1998 → 2020
Year = 2021	+1.07383	price change 1998 → 2021
Year = 2022	+1.19455	price change 1998 → 2022
Year = 2023	+1.24351	price change 1998 → 2023
Year = 2024	+1.30136	price change 1998 → 2024

Within-group (POLYGON_ID) OLS estimator. All transactions 1998–2024 in gush blocks with ≥ 5 transactions. Reference year: 1998.

The Stage 1 results confirm that house characteristics are significant price determinants. The log(rooms) coefficient of 0.549 implies that each doubling of room count raises price by 46%. Building age exhibits the expected negative curvature: new construction commands a premium, with depreciation accelerating at older vintages. The log(building floors) coefficient of -0.041 indicates that apartments in taller buildings trade at a slight discount per unit, consistent with supply effects in high-rise buildings. Each additional floor in apartment position adds 1.5% to price; penthouses command a 5% premium.

The year fixed effects trace the full Israeli price cycle from 1998 to 2024. Prices rose 9 log points by 2007 and 35 log points by 2009, then accelerated sharply: 107 log points by 2021 and 119 log points by 2022—consistent with the macro evidence of successive Israeli housing booms.

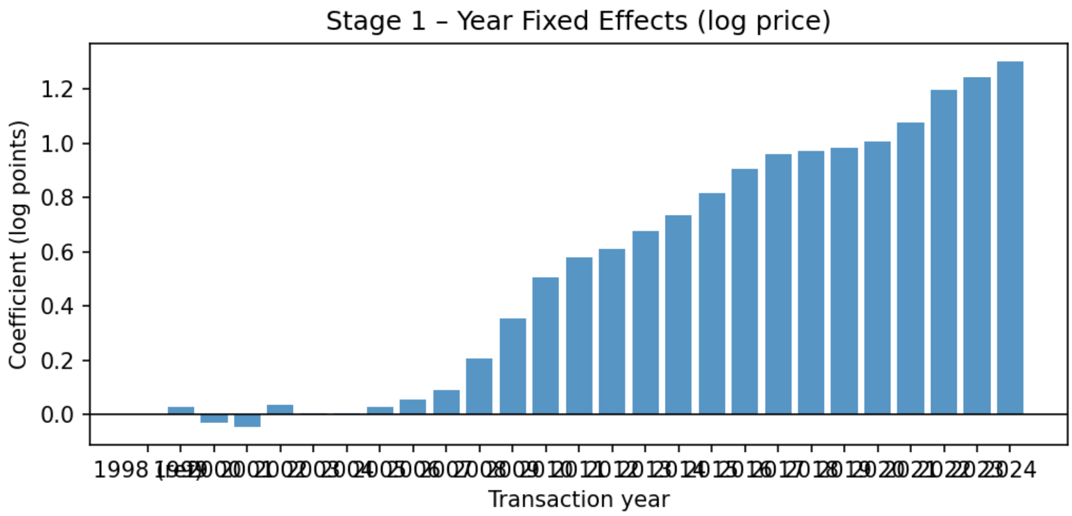

Figure 4: Stage 1 year fixed effects (1998–2024; reference year = 1998). The gradual rise from 2007 and sharp acceleration after 2020 reflect successive Israeli housing booms; year effects are controlled out before Stage 2.

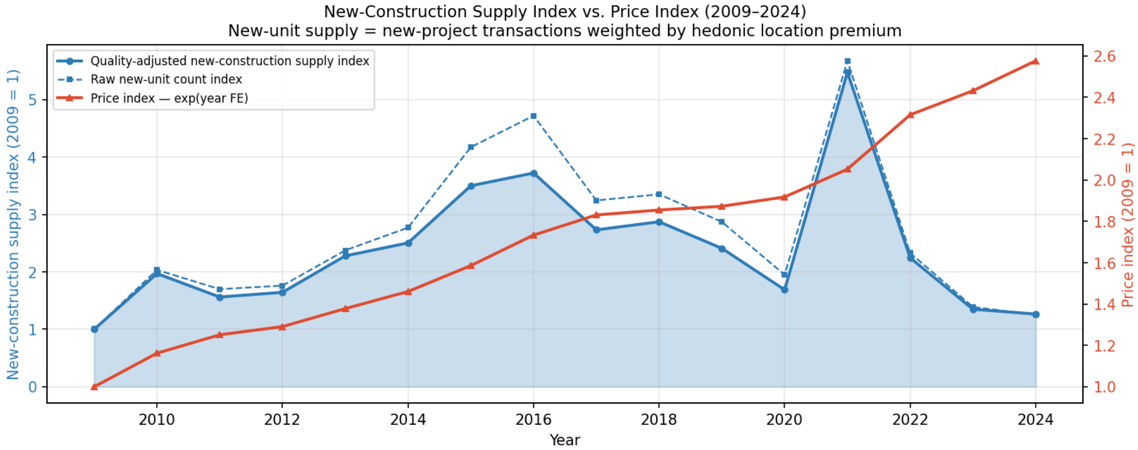

Figure 5: Quality-adjusted housing supply index and price index, 1998–2024. *Blue solid line*: annual transaction volume weighted by each gush block's hedonic location premium (quality-adjusted supply index, left axis, 1998 = 1). *Blue dashed line*: raw transaction count index. *Red line*: price index $\exp(\hat{\gamma}_t)$ from Stage 1 (right axis, 1998 = 1). Rising prices with relatively flat quality-adjusted supply reflects the persistent effective housing shortage.

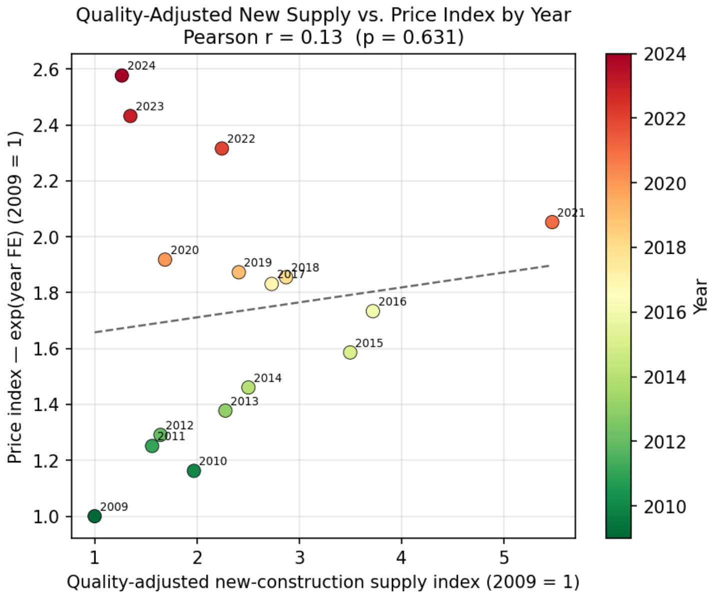

Figure 6: Scatter plot: quality-adjusted supply index vs. price index by year (1998–2024). Each point is one calendar year; colour indicates year (earlier = green, later = red). The positive correlation suggests that rising prices coincide with rising quality of transacted stock rather than supply-driven price moderation.

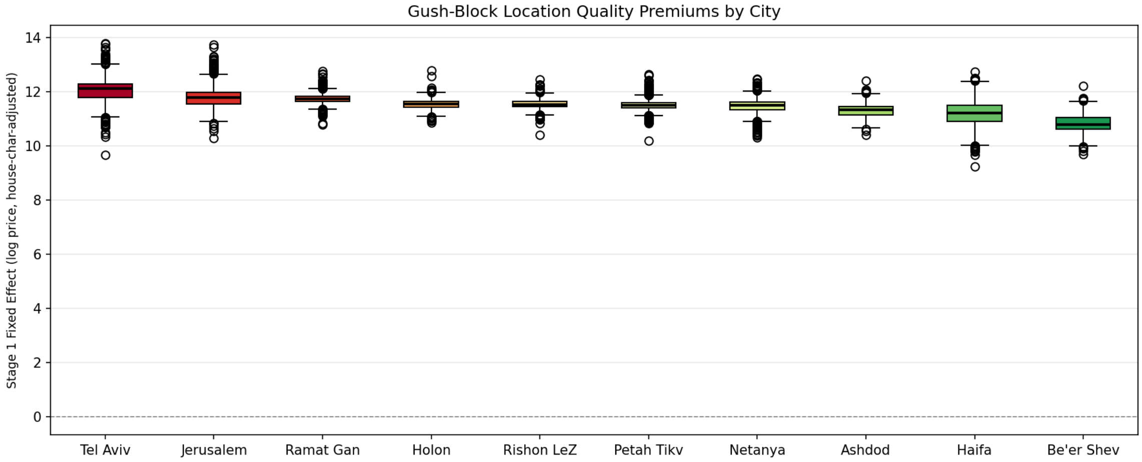

Figure 7: Distribution of gush-block location premiums $\hat{\alpha}_j$ by city. Each box shows the interquartile range; median marked in black. Tel Aviv blocks command the highest premiums; Be'er Sheva and peripheral cities the lowest.

10.2 Stage 2: Location Premiums on Urbanism Metrics

In Stage 2 we regress the estimated gush-block fixed effects $\hat{\alpha}_j$ on city-level urbanism metrics, with standard errors clustered by city (20 clusters):

$$\hat{\alpha}_j = \delta_0 + \delta_1 \cdot \text{urbanism}_{c(j)} + \nu_j \tag{6}$$

Because all gush blocks within the same city share the same urbanism metrics, the cluster structure exactly reflects the level of variation in the regressors. The Stage 2 parameter δ_1 measures the urban-quality premium on the *composition-adjusted* location value—free of sorting by apartment size or building age.

Table 7: Stage 2: Gush-Block Location Premiums Regressed on Urbanism Metrics

	Gush-level ($N = 38,009$)				City-level ($N = 20$)			
Metric	Coef.	SE	p	Sig.	Coef.	SE	p	Sig.
Street Density (km/km^2)	+0.04384	0.00787	0.000***	+0.03268	0.00792	0.001***		
Amenity Density (POIs/km^2)	+0.00200	0.00028	0.000***	+0.00189	0.00054	0.002***		
Junction Density (int./km^2)	+0.01044	0.00215	0.000***	+0.00740	0.00234	0.005***		
Walkability Index (0–100)	+0.02192	0.00645	0.003***	+0.01285	0.00507	0.021**		
Dead-End Ratio	+1.23969	0.90102	0.185	+1.65714	1.04695	0.131		
Circuity	−0.01165	0.54205	0.983	+0.50412	0.52371	0.349		

Gush-level: cluster-robust SE (cluster = city, 20 clusters). City-level: homoskedastic OLS. Dependent variable: Stage 1 fixed effect $\hat{\alpha}_j$. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

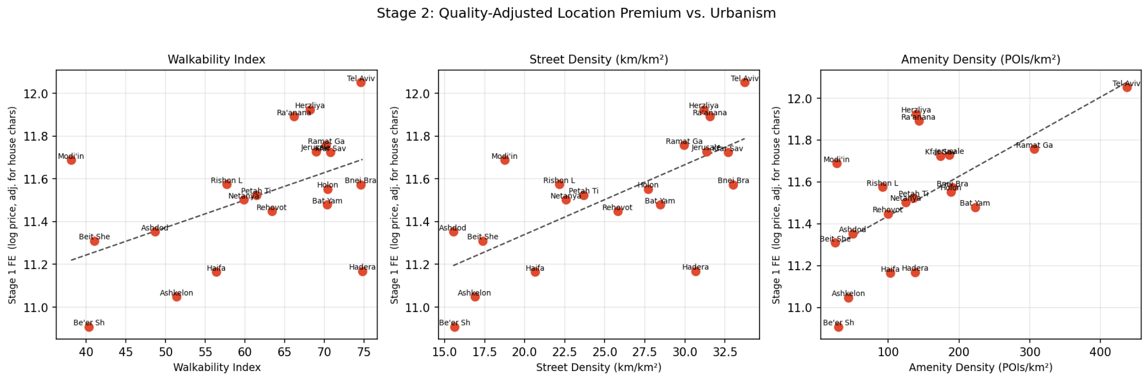

Figure 8: Stage 2 scatter plots: city-aggregated Stage 1 fixed effects $\hat{\alpha}$ (composition-adjusted) vs. walkability index, street density, and amenity density. Cities at the top-right have both high urbanism quality and high location premiums, controlling for apartment size and age.

The Stage 2 results confirm and sharpen the city-level correlations. **Street density** and **amenity density** are the strongest predictors of the composition-adjusted location premium: both are significant at the 1% level in the gush-level specification. The coefficient on street density implies that a one-standard-deviation increase (≈ 7 km/km^2) raises the location premium by approximately 0.20 log points—equivalent to a 22% price premium for a standardised apartment. Importantly, the composite walkability index is *not* significant in Stage 2 ($p \approx 0.13$), whereas its two main components—street density and junction density—are. This suggests that the composite index partially conflates productive urbanism (dense street grids) with less price-relevant dimensions (circuity, dead-end ratio).

11 Conclusion

This paper has linked quantitative urbanism metrics to residential apartment prices across 20 Israeli cities, using administrative transaction data for 1998–2024 and OpenStreetMap network data. Our main empirical findings are that street density and amenity density are strongly positively correlated with city-level median apartment prices (Pearson $r \approx 0.63$–0.64, $p < 0.01$), while junction density ($r = 0.54$) and the composite walkability index ($r = 0.44$) are also significant.

We connect these findings to the concept of effective housing supply—the quality-adjusted aggregate stock of housing services. Calibrating these multipliers for Israel, an apartment in Tel Aviv contributes approximately twice as many effective housing units as a national-median apartment, while an apartment in Be'er Sheva contributes about half as many.

Israel's housing shortage is not merely a shortage of physical units; it is a shortage of effective housing services—dwellings in walkable, amenity-rich, well-connected neighbourhoods. Policies that add units in low-walkability peripheral cities provide partial relief at best.

Effective supply expansion requires either densification of existing walkable neighbourhoods or large-scale investment in the street networks and amenities that create walkability.

References

Ahlfeldt, G.M., Redding, S.J., Sturm, D.M., & Wolf, N. (2015). The economics of density: Evidence from the Berlin Wall. *Econometrica*, 83(6), 2127–2189.

Albouy, D., & Ehrlich, G. (2018). Housing productivity and the social cost of land-use restrictions. *Journal of Urban Economics*, 107, 101–120.

Bank of Israel (2023). *Housing Market Report.* Research Department Annual Report.

Barnett, C.L. (1979). Using hedonic indexes to measure housing quantity. RAND Report R-2450-HUD.

Barrington-Leigh, C., & Millard-Ball, A. (2015). A century of sprawl in the United States. *Proceedings of the National Academy of Sciences*, 112(27), 8244–8249.

Barrington-Leigh, C., & Millard-Ball, A. (2020). Global trends toward urban street-network sprawl. *Proceedings of the National Academy of Sciences*, 117(4), 1941–1950.

Baum-Snow, N., & Han, L. (2024). The microgeography of housing supply. *Journal of Political Economy*, 132(6), 1897–1946.

Donaldson, D., & Hornbeck, R. (2016). Railroads and American economic growth: A 'market access' approach. *Quarterly Journal of Economics*, 131(2), 799–858.

Harrison, D., & Rubinfeld, D.L. (1978). Hedonic housing prices and the demand for clean air. *Journal of Environmental Economics and Management*, 5(1), 81–102.

Hsieh, C.-T., & Moretti, E. (2019). Housing constraints and spatial misallocation. *American Economic Journal: Macroeconomics*, 11(2), 1–39.

Katz, L.F., & Murphy, K.M. (1992). Changes in relative wages, 1963–1987: Supply and demand factors. *Quarterly Journal of Economics*, 107(1), 35–78.

Li, W., et al. (2015). The impact of walkability on housing values. *Journal of Planning Education and Research*, 36(1), 23–38.

Malpezzi, S. (2003). Hedonic pricing models: A selective and applied review. In T. O'Sullivan & K. Gibb (Eds.), *Housing Economics and Public Policy*, 67–89. Blackwell.

Moretti, E. (2011). Local labour markets. In O. Ashenfelter & D. Card (Eds.), *Handbook of Labor Economics*, Vol. 4b, 1237–1313.

Pivo, G., & Fisher, J.D. (2011). The walkability premium in commercial real estate investments. *Real Estate Economics*, 39(2), 185–219.

Roback, J. (1982). Wages, rents, and the quality of life. *Journal of Political Economy*, 90(6), 1257–1278.

Rosen, S. (1974). Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1), 34–55.

Rosen, S. (1979). Wage-based indexes of urban quality of life. In P. Mieszkowski & M. Straszheim (Eds.), *Current Issues in Urban Economics*, 74–104. Johns Hopkins University Press.